

IR Based Automatic Hand Sanitiser Dispenser

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This project is based on an infrared (IR) sensor, which detects the presence of hands and consequently activates the motor pump to dispense alcohol-based sanitiser. The circuit is economical, reliable, and can be easily constructed, as explained below.

Circuit and working

The block diagram of IR based automatic hand sanitiser dispenser is shown in Fig. 1, and its circuit diagram is shown in Fig. 2. It consists of an IR sensor module (IR1), transistor BC547 (T1), 5V single-changeover relay (RL1), diode 1N4007 (D1), LED1, mini submersible DC motor pump, and a few other components.

Note that there are two types of IR modules commonly available in the market. One is based on LM393, whose output goes low when an object is detected, and another is based on LM358, whose output goes high when an object is detected. This project uses a 3-pin IR module based on LM358.

The project uses 5V DC to drive the circuit. When the IR sensor detects a hand, its output goes high. This high

output drives relay driver transistor (BC547) and energises relay RL1. Submersible DC pump gets power supply through NO contacts of the relay. This turns on the DC pump, and hand sanitiser liquid starts to flow out of the small plastic pipe fitted on the bottle.

When there is no hand or object in front of the IR sensor, its output goes low. This low output cuts off relay driver transistor (BC547) and

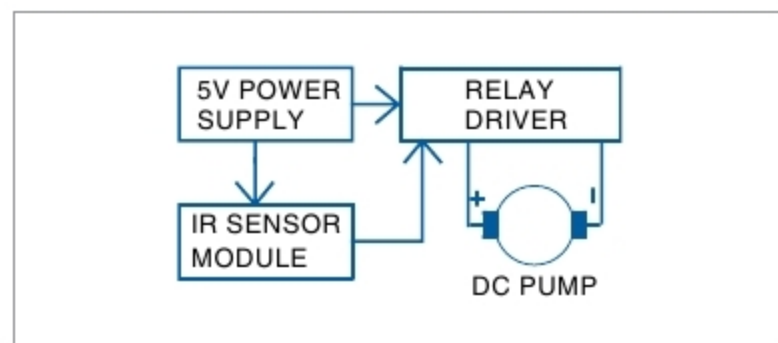


Fig. 1: Block diagram of IR based automatic hand sanitiser dispenser

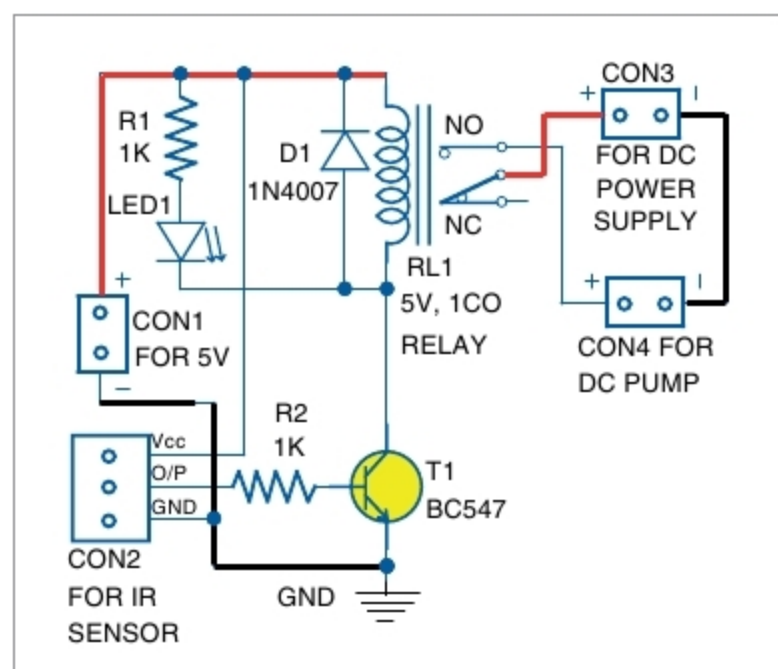


Fig. 2: Circuit diagram of the IR based automatic hand sanitiser dispenser

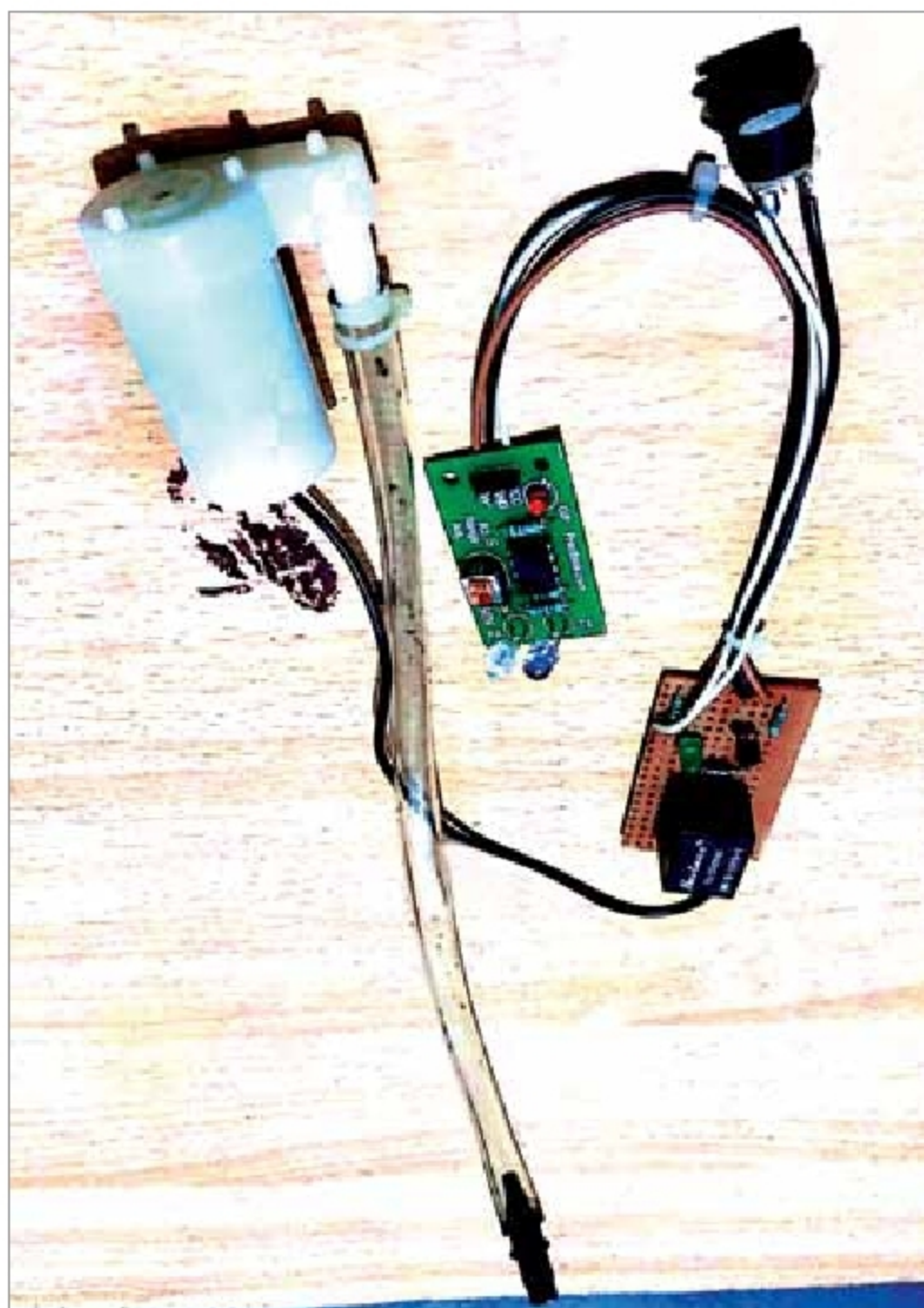


Fig. 3: Various components used in the project